Exhibit 3: Effect of Trimming and Winsorizing

	Arithmetic Mean	Trimmed Mean (Trimmed 5%)	Winsorized Mean (95%)
Mean	0.035%	0.048%	0.038%
Number of Observations	1,258	1,194	1,258

The trimmed mean eliminates the lowest 2.5 percent of returns, which in this sample is any daily return less than −1.934 percent, and it eliminates the highest 2.5 percent, which in this sample is any daily return greater than 1.671 percent. The result of this trimming is that the mean is calculated using 1,194 observations instead of the original sample's 1,258 observations.

The winsorized mean substitutes a return of −1.934 percent (the 2.5 percentile value) for any observation below −1.934 and substitutes a return of 1.671 percent (the 97.5 percentile value) for any observation above 1.671. The result in this case is that the trimmed and winsorized means are higher than the arithmetic mean, suggesting the potential evidence of significant negative returns in the observed daily return distribution.

Measures of Location

Having discussed measures of central tendency, we now examine an approach to describing the location of data that involves identifying values at or below which specified proportions of the data lie. For example, establishing that 25 percent, 50 percent, and 75 percent of the annual returns on a portfolio provides concise information about the distribution of portfolio returns. Statisticians use the word **quantile** as the most general term for a value at or below which a stated fraction of the data lies. In the following section, we describe the most commonly used quantiles—quartiles, quintiles, deciles, and percentiles—and their application in investments.

Quartiles, Quintiles, Deciles, and Percentiles

We know that the median divides a distribution of data in half. We can define other dividing lines that split the distribution into smaller sizes. **Quartiles** divide the distribution into quarters, **quintiles** into fifths, **deciles** into tenths, and **percentiles** into hundredths. The **interquartile range** (IQR) is the difference between the third quartile and the first quartile, or IQR = $Q_3 - Q_1$.

Example 2 illustrates the calculation of various quantiles for the daily return on the EAA Equity Index.

EXAMPLE 2

Percentiles, Quintiles, and Quartiles for the EAA Equity Index

Using the daily returns on the EAA Equity Index over the past five years and ranking them from lowest to highest daily return, we show the return bins from 1 (the lowest 5 percent) to 20 (the highest 5 percent) as follows: